

Randy Ren

+1 825-431-3285 | me@randyren.org | randyren.org | github.com/randyyren278

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, TypeScript, $\text{\LaTeX}$, System Verilog
Tools: Jira, Confluence, GitHub, Fusion360, Google Cloud Platform, Figma, Blender, Microsoft Office Suite
Hardware: NVIDIA Jetson Orin, Raspberry Pi, Arduino, DE1-SoC

EXPERIENCE

Machine Learning Intern

September 2025 – December 2025

Ledcor Technical Services

Vancouver, BC

- Automated quality control system using **YOLO** object detection and **vision language models**, streamlining infrastructure installations and saving **33+ days** annually in manual review time.
- Developed end-to-end ML pipeline in **Python** trained on historical audit datasets achieving **99%** agreement with human decision making.

Project Management Intern

May 2025 – August 2025

Ledcor

Vancouver, BC

- Coordinated **Quality Assurance** workflows by scheduling tests, monitoring field inspections, and overseeing full project lifecycle across **hundreds of fiber optic installation sites**.
- Provided technical and administrative support to superintendents, design engineers, and build partners, ensuring consistent communication and documentation flow.
- Providing cross-functional support in project coordination and data analysis contributing to the exploration of **machine learning** and **AI** applications in construction workflows.

Software Engineering Intern

May 2024 – August 2024

Visa

San Francisco, CA

- Prototyped personalized authorization models for payment anomaly detection, utilizing **Python** and **scikit-learn** to explore novel approaches to fraud prevention.
- Contributed to backend microservices powering Visa's real-time authorization flow and **Smarter Stand-In Processing (STIP)** system.
- Developed automated testing frameworks in **Java** using **JUnit** and **Mockito**, validating AI model predictions against historical bank authorization decisions.

PROJECTS

Facial Recognition System | GitHub

- Developed a facial recognition system using **Python** and **OpenCV** for real-time image and video processing.
- Implemented a facial recognition library for robust face detection and recognition, achieving **95%** accuracy in facial identification.

Personal Portfolio Website | GitHub |

- Designed and developed a fully responsive portfolio site using **Next.js**, **Tailwind CSS**, and **Framer Motion** to showcase projects, resume, and contact information.
- Built immersive 3D interactive world using **Three.js** and **React Three Fiber**, implementing custom shaders, lighting systems, and camera controls for engaging user experience across mobile and desktop platforms.

EXTRACURRICULAR

Competitive Track & Field Athlete

2019 – Present

U of C Athletics Club and UBC Thunderbirds Endurance

Calgary & Vancouver, Canada

- Took on **leadership responsibilities** helping plan and execute team workouts among senior athletes.
- Personal bests: **51.14** (400m), **15:17** (5km), **3:15.39** (4x400m relay).
- Competed in numerous **high-level track meets**, including **Athletics Canada Nationals**.

EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Applied Science in Electrical Engineering

Sep. 2023 – Apr. 2028